

PASSIVE STT-MRAM CROSSBARS FOR IN-MEMORY HARDWARE-IN-THE-LOOP AND CONTINUAL LEARNING

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In-memory computing has emerged as a promising solution, particularly through memristive crossbar arrays, capable of performing multiply-and-accumulate operations directly within memory. This work introduces hardware-in-the-loop learning using passive 15X15 STT-MRAM crossbars. We focus on different scenarios, either using discrete (quantized) or continuous (full precision) weights encoded with one (binary) or two (ternary) Magnetic Tunnel Junctions, to evaluate the impact of the accuracy on the subsequent learning process. Additionally, we explore continual learning by implementing metaplasticity and demonstrate its hardware implementation with two sequential single tasks. These results highlight the potential of MRAM-based in-memory architectures for adaptive, energy-efficient, and scalable AI hardware.

Index Terms—In-Memory-Computing, MRAM, continual learning, hardware in the loop

I. INTRODUCTION

Deep neural networks (DNNs) have significantly propelled the advancement of artificial intelligence but often demand substantial energy and computational resources [1], largely due to the limitations of traditional von Neumann architectures. In-memory computing offers a promising alternative by co-locating computation with memory, reducing data movement and improving efficiency [2]. Memristive crossbar arrays are especially effective in this paradigm [3], performing multiply-and-accumulate (MAC) operations efficiently by exploiting Kirchhoff's law (Figure 1). While conventional CMOS memories offer some advantages, emerging non-volatile memory (NVM) technologies such as MRAM, RRAM, and PCM provide more favorable tradeoffs in terms of performance and energy efficiency. Among these, magnetic tunnel junctions (MTJs), which form the core of MRAM, stand out for their binary switching behavior, low power consumption, and high endurance, making them well-suited for neuromorphic computing. MRAM technology is particularly compatible with binary neural networks (BNNs), which use binary weights and logic-based computation to minimize power and memory requirements.

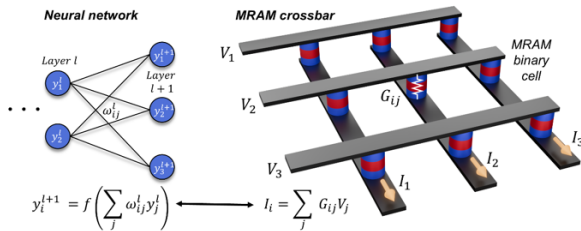


Figure 1. MRAM memristive crossbar. Example of MAC operation in a neural network and its equivalent on a 3×3 crossbar architecture.

Although prior works have demonstrated inference capabilities with MRAM-based crossbars [4, 5, 6], learning directly on such hardware remains relatively unexplored. This work introduces one of the first demonstrations of both hardware-in-the-loop learning and continual learning using passive 15×15 STT-MRAM crossbars.

II. EXPERIMENTS

We have used a spin-transfer torque (STT)-MRAM crossbar integrating magnetic tunnel junctions (MTJs) nanopillars with a 25 nm critical diameter. The array consists of a 15×15 matrix of individual MTJs, each composed of a CoFeB based fixed layer and a CoFe/CoFeB free layer separated by an MgO tunnel barrier. The crossbar characterization and programming are performed using a specific PCB customized for crossbar measurements, the LOTUS platform [7].

III. RESULTS

Hardware-in-the-loop (HIL) training has been implemented, in which the physical crossbar actively participates in every weight update cycle. We focus on different scenarios, either using discrete (quantized) or continuous (full precision) hidden weights encoded with one (binary) or two (ternary) MTJs, to evaluate the impact of the accuracy of the subsequent learning process. In Figure 2, the statistics of 125 runs are shown for different programming conditions and Neural Network sizes. Using real hidden weights (in blue), in either differential or single mode, the median accuracy reaches 100 %, which demonstrates that the neural network is able to learn from a random configuration. The dispersion is slightly smaller for the differential case. The discrete hidden weights (in red) show a loss of accuracy, especially drastic for the single programming

of small size but it can be recovered by increasing the number of MTJs.

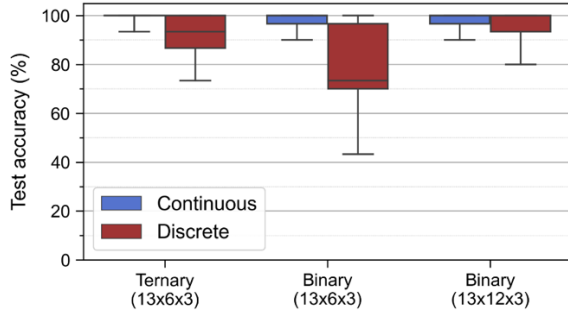


Fig. 2. Distributions of test accuracy over 125 runs for different programming schemes on a simple task (ECG). The extremities of each boxplot correspond to the 10 and 90 percentiles. In ternary mode (resp. binary), two MTJs (resp. one) are used to encode a weight.

Additionally, in order to implement continual learning, we explore the ability of our reduced network to learn two tasks sequentially (Wine and ECG) for differential and single programming. We implement a metaplastic algorithm with real hidden weights, inspired by [8]. In Figure 3, a typical example of accuracy trend with differential programming are given for a metaplastic parameter of $m = 2.5$. The second task, ECG, is learned from epoch 150. A slight drop is visible but the network continues to retain some accuracy even as the second task is learned. Due to the small size of the network, the retention of the first task is not perfect but it could be improved by using more synapses.

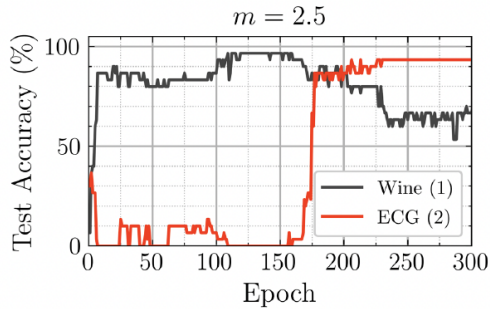


Fig. 3. Continual learning hardware implementation : test accuracy evolution over 300 epochs for a metaplastic parameter $m = 2.5$ in differential mode showing that the arrays is able to learn 2 tasks sequentially (Wine and ECG)

IV. CONCLUSION

In summary, we have demonstrated hardware-in-the-loop training and continual learning using a 15x15 STT-MRAM crossbar array, establishing a complete experimental framework in which the memory hardware actively participates in the learning process. Scaling to larger crossbars would allow the implementation of wider and deeper networks, which are expected to significantly improve classification accuracy. This work thus demonstrates that STT-MRAM crossbar arrays can evolve from inference-only platforms to adaptive neuromorphic platforms capable of continual learning for edge AI.

References

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